THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Water is a compound and not a mixture of hydrogen and oxygen. Reason (R): Hydrogen and oxygen in water are chemically combined in a fixed ratio. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. 4 g of hydrogen combines with 32 g of oxygen to form water. The mass of water formed and the ratio of hydrogen to oxygen are: 1

(a) 28 g, 1 : 8 **(b)** 36 g, 1 : 4 **(c)** 36 g, 8 : 1 **(d)** 36 g, 1 : 8

3. A coin is made of an alloy of copper and nickel. This coin is a: 1

(a) compound **(b)** heterogeneous mixture
(c) homogeneous mixture **(d)** element

4. Assertion (A): Brass is a mixture of copper and zinc. Reason (R): The proportions of copper and zinc in brass can vary. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. 9 g of water is decomposed. The masses of hydrogen and oxygen obtained are: 1

(a) 4.5 g and 4.5 g **(b)** 2 g and 7 g **(c)** 8 g and 1 g **(d)** 1 g and 8 g

6. Zinc reacts with dilute acid and produces: 1

(a) oxygen gas **(b)** chlorine gas **(c)** nitrogen gas **(d)** hydrogen gas

7. Assertion (A): Sodium is stored under kerosene. Reason (R): Sodium is a very reactive metal. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. The total number of atoms in 2 molecules of H_2SO_4 is: 1

(a) 14 **(b)** 4 **(c)** 28 **(d)** 7

9. A shiny, hard element conducts electricity and turns black in air with sulphur compounds. It could be: 1

(a) silver **(b)** sulphur **(c)** neon **(d)** carbon

10. Assertion (A): Graphite conducts electricity although carbon is a non-metal. Reason (R): Graphite is an exception among non-metals. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. 4 g of hydrogen reacts completely with oxygen forming 36 g of water. Find the mass of oxygen and the mass ratio of hydrogen to oxygen. 2

12. Explain why alloys like solder are used to join metals. 2

13. Why are noble gases unreactive and how is this useful? 2

14. How many atoms are present in 3 molecules of CO_2? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A student has a sample of a grey powder. It is attracted by a magnet and gives hydrogen when acid is added. Identify the powder as an element, compound or mixture and explain. 3

16. Explain why diamonds do not conduct electricity but graphite does. 3

17. A fruit juice is made of water, sugar, acids and colour. Describe its nature and how you may separate water from it. 3

OR

Compare the properties of sodium, chlorine and sodium chloride to show that a compound is different from its elements. 3

18. Why does a solution of salt and sugar in water form a homogeneous mixture of three components? 3

OR

How can you prove that carbon is present in sugar? State the observation and inference. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Design an investigation to distinguish between a mixture and a compound using iron and sulphur. Write the procedure, the observations before and after heating and the conclusion. 5

OR

Explain the particle picture of an element, compound and mixture with the example of oxygen, water and air, and the numbers of atoms. 5

20. Explain the law of constant proportions with calculations for water (1 : 8 by mass), and find the masses of hydrogen and oxygen in 45 g of water. Also explain what the law shows about compounds from any source. 5

OR

Discuss the chemical test and physical test used to distinguish metals and non-metals, including reaction with oxygen and acid. 5

END OF QUESTION PAPER — Total: 40 Marks